

QA AGENT

High-Level Client Requirements

AI-Assisted Quality Engineering

1. Requirement Overview

The QA Agent is expected to provide an AI-assisted quality engineering capability that can understand an application through application exploration, validate new or modified functionality, execute tests using natural-language instructions, generate automation when required, maintain existing automation, and provide scheduled and actionable test results.

2. Core Client Requirement

The QA Agent should be capable of independently understanding and learning an application through application exploration using a provided URL and credentials, without depending on detailed functional documentation, wireframes, or continuous SME support. When new or modified functionality is introduced, the agent should use the existing application knowledge together with the available feature information to identify relevant test coverage, execute appropriate validations, generate automation when required, and provide actionable results. The agent should also support natural-language, on-demand test execution and scheduled daily sanity testing with automated reporting.

3. High-Level Requirements

ID	Requirement	High-Level Description
QA-01	AI-Based Application Discovery & Understanding	Explore the application using the provided URL and valid credentials and identify modules, screens, navigation paths, user journeys, validations, and observable business behaviour.
QA-02	Business Flow Identification	Infer existing business flows and relationships from application exploration and available high-level business information, without requiring detailed flow documentation or continuous SME support.
QA-03	Application Knowledge Base	Maintain application knowledge acquired through discovery, business information, feature changes, execution results, and approved automation.
QA-04	New Feature / Change Validation	Use a basic feature document or description to understand a new or modified feature in the context of the existing application, identify relevant areas, execute validation, and provide results.
QA-05	Natural-Language Test Execution	Accept plain-English instructions to execute sanity, regression, specific business flows, feature-specific testing, or other supported validations.
QA-06	On-Demand Automation Generation	Generate reusable automation scripts for identified or executed flows when required, with minimal manual scripting.

QA-07	Self-Healing / Adaptive Automation	Identify supported UI, locator, navigation, and application changes and provide AI-assisted healing or adaptation of existing automation where technically feasible.
QA-08	Auto-Correction / Flow Regeneration	Identify affected workflows after application changes and adapt or regenerate relevant automation where technically feasible.
QA-09	Change Impact Analysis	Identify existing flows and automation assets potentially affected by application changes and recommend or execute relevant regression coverage.
QA-10	Automated Daily Sanity Testing	Support a task scheduler for configured daily sanity execution.
QA-11	Automated Reporting & Notifications	Automatically generate execution results and distribute them through the client's agreed notification or reporting channel, such as email.
QA-12	AI Test Analytics & Insights	Provide actionable insights beyond pass/fail results, including failure analysis, recurring failures, regression trends, coverage, and other supported metrics.
QA-13	Continuous Application Learning	Progressively improve application understanding as additional business information, feature changes, discoveries, execution results, and approved automation become available.

4. Expected User Experience

- User provides application URL and credentials.
- QA Agent explores and builds application understanding.
- Agent identifies and maintains relevant business flows.
- User provides a basic document or description when a new or modified feature is introduced.
- Agent identifies relevant test coverage and executes validation.
- Agent provides results and evidence.
- User can request automation generation when required.
- Configured scheduler can execute daily sanity testing.
- Results can be automatically distributed through the agreed reporting channel.

5. Natural-Language Execution Examples

Example User Instruction	Expected Agent Behaviour
"Run the sanity test for the application."	Identify configured sanity coverage, execute it, and return results.
"Run regression for the order flow."	Identify relevant regression coverage and execute

	supported flows.
"Test the new product search feature."	Use feature information and existing application knowledge to identify and execute relevant validation.
"Create automation for this flow."	Generate automation assets for the identified or executed flow, subject to supported capabilities.

6. Client Inputs & Dependencies

The client is expected to provide the minimum information and access required for the agent to understand and validate the application. Detailed functional documentation, wireframes, and continuous SME support are not assumed to be mandatory inputs.

- Application URL and valid credentials.
- High-level business/application overview where available.
- Basic feature information or documentation for new or modified functionality.
- Required test data and environment access, where applicable.
- Access and permissions necessary to execute the agreed testing scope.
- Agreed reporting/notification destination for scheduled execution results.

7. Coverage & Capability Dependency

The depth and accuracy of automation and validation coverage will depend on application accessibility, available test data, feature complexity, environment stability, and the quality of information available to the agent. Coverage is expected to expand progressively as the agent gains additional application knowledge.

8. High-Level Success Criteria

- Application can be explored and understood using URL and credentials.
- Relevant existing business flows can be identified without requiring detailed flow documentation.
- New or modified features can be validated using basic feature information.
- Users can initiate supported test flows using natural-language instructions.
- Automation can be generated when requested.
- Supported application changes can be handled through AI-assisted healing or flow adaptation.
- Daily sanity execution can be scheduled.
- Execution results can be automatically reported through the agreed channel.
- Application knowledge and automation coverage can progressively improve over time.

9. Scope Clarification

This document captures the client's high-level expectations for the QA Agent. Detailed technical design, supported application boundaries, implementation approach, measurable service levels, and final acceptance criteria should be defined during detailed solution design and proof-of-concept validation.